After trading costs, how much money remains?

Keep the one-day return rule from Week 4. Separate cost rates, execution dates and the account after costs; do not select weighting methods.

Issue	Question
01	Can daily prices reveal the cost of our trade?
02	How many dollars does 4 basis points cost?
03	How do three commission schedules charge for the same stock?
04	At USD 185.14, how much extra would an immediate purchase cost?
05	When is the spread counted twice?
06	Record the purchase cost on the 9th or the 10th?
07	Does a purchase and a sale each incur a cost?
08	First round trip: how much of USD 1,000 remains?
09	Why subtract 4 basis points to obtain the net return?
10	Why does the terminal wealth gap differ from total fees?
11	How can USD 1,000 cover both stock and purchase cost?
12	How does buying one close later change the first holding return?
13	Can the first worse return determine the entire delayed account?
14	Do fewer trades necessarily leave more money?
15	What happens to terminal wealth when only the cost rate rises?
16	Which cost rate eliminates this historical profit?
17	How do cash, buy-and-hold and the trading rule compare?
18	With a USD 1 minimum per order, does a small order still cost only 4bp?

Which costs are added to the same trading rule?

AAPL, 2024/1/2 to 6/28. Request stock when the Close exceeds the previous trading-day Close; otherwise request cash. Trade at the next trading-day Close.

Start with USD 1,000 cash and allow fractional shares. Exclude dividends, taxes and cash interest. At the final Close, value the account without a new trade or forced sale. Next-close execution is a teaching assumption, not a restriction on U.S. trading.

Predict: does the stock request on 1/8 execute on 1/8 or at the 1/9 Close?

```
import numpy as np
import pandas as pd
import yfinance as yf
aapl = yf.Ticker("AAPL").history(
    start="2024-01-02", end="2024-07-01", interval="1d",
    auto_adjust=False, actions=True)
aapl.index = aapl.index.strftime("%Y-%m-%d")
close = aapl["Close"]
assert close.index.is_unique and close.index.is_monotonic_increasing
assert np.isfinite(close).all() and (close > 0).all()
price_change = close / close.shift(1) - 1
signal = price_change.gt(0).astype(float).where(price_change.notna())
dates = ["2024-01-08", "2024-01-09", "2024-01-10"]
pd.DataFrame({"Observation date": dates,
    "Close (USD/share)": [f"{close[d]:.2f}" for d in dates],
    "After-close request": ["Hold stock" if signal[d] == 1 else "Cash" for d in dates]})
```

Observation date	Close (USD/share)	After-close request
2024-01-08	185.56	Hold stock
2024-01-09	185.14	Cash
2024-01-10	186.19	Hold stock

signal is the after-close request: 1 for stock, 0 for cash. The 1/8 stock request buys at the 1/9 Close; the 1/9 cash request sells at the 1/10 Close. The 124 prices form 123 intervals. The first date has no previous Close; the account starts in cash.

The 4 bp rate per direction is assumed, not estimated from daily prices. Approximate dollar cost by rate times the change in stock weight times opening account value. See Issue 2 for basis points and Issue 11 for the difference from an exact shares-and-cash budget.

ISSUE 01

Can daily prices reveal the cost of our trade?

Separate market prices from order records. A price change alone does not show how much extra an order cost.

Predict: will this table include our order size, execution time and commission?

```
dates = ["2024-01-08", "2024-01-09"]
pd.DataFrame({"Observation date": dates,
    "Open (USD/share)": aapl.loc[dates, "Open"].round(2).values,
    "Close (USD/share)": close.loc[dates].round(2).values})
```

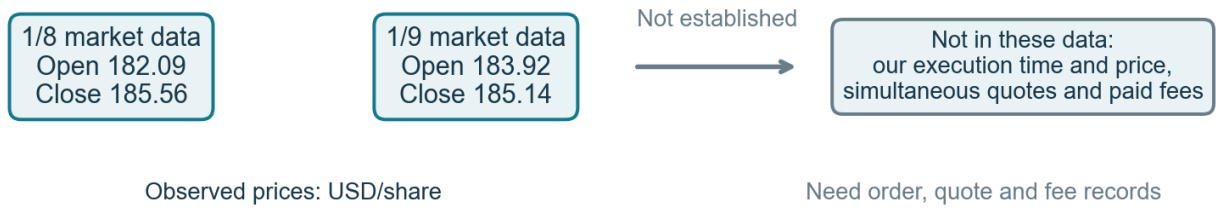
Observation date	Open (USD/share)	Close (USD/share)
2024-01-08	182.09	185.56
2024-01-09	183.92	185.14

The 1/8 Close of 185.56 forms the buy signal; the main example buys at the 9th Close of 185.14. The 9th Open of 183.92 and Close of 185.14 are observations at different times. Their difference is not automatically this order's slippage.

Commission requires a fee schedule; spread requires simultaneous bid and ask quotes. Slippage compares our execution price with a specified reference, such as the midpoint at order submission. This needs our execution record, which daily data do not contain.

Use an assumed 4 bp per direction to study the account effect. These daily prices do not establish actual spreads, slippage or execution prices.

Daily prices are not our execution or cost records



Open-to-Close movement is not this order's slippage. These data provide neither our execution nor simultaneous quotes.

Discuss

To measure slippage for the 1/9 purchase, which execution record is missing from the daily price table?

Sources: [1], [2], [3], [4].

ISSUE 02

How many dollars does 4 basis points cost?

Convert the rate to a decimal, then multiply by the amount traded. A 4 bp rate does not mean charging the whole account every day.

1 basis point (bp) is 0.01 percentage points, or 0.01%. Thus 4 bp = 0.04% = 0.0004. The 1/9 Close is about USD 185.14/share. Assume buying one share at that price solely to calculate the fee.

q is shares, C is price in USD/share, F is cost in USD, and c is the decimal rate. For one share, q=1 and the traded amount is USD 1×185.14.

$$F = cqC, \qquad 0.0004(1)(185.14) \approx 0.074056 \text{ USD}$$

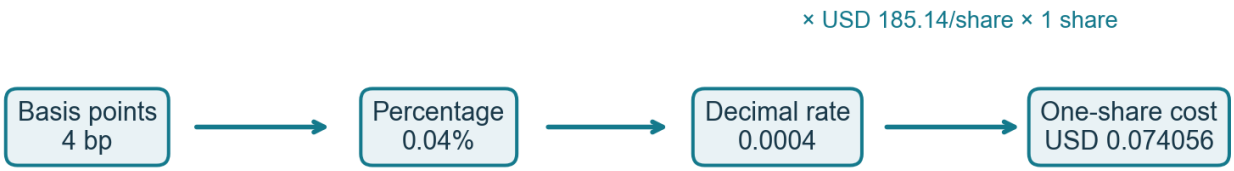
Predict: is the 4 bp cost for one share above or below one cent?

```
price = close["2024-01-09"]
bp = np.array([1, 4, 10, 100])
pd.DataFrame({"Assumed rate (bp)": bp, "Rate (%)": bp/100,
    "One-share fee (USD)": (price*bp/10000).round(5)})
```

Assumed rate (bp)	Rate (%)	One-share fee (USD)
1	0.01	0.01851
4	0.04	0.07406
10	0.1	0.18514
100	1.0	1.8514

4 bp costs about USD 0.07406, or 7.41 cents, not 4%. If only part of the capital trades, charge only that traded amount.

One share on 2024/1/9: convert 4 bp to a decimal first



4 bp is not 4%. At the 1/9 Close, the assumed one-share cost is about USD 0.07406.

Discuss

At the same price, what is the 4 bp dollar cost for ten shares?

Sources: [2], [5].

ISSUE 03

How do three commission schedules charge for the same stock?

A fixed fee depends on order count, a per-share fee on shares, and a proportional fee on traded value. Identify the schedule before comparing costs.

Use the 1/9 price of USD 185.14/share. Assume one order for 1, 10 or 100 shares. Compare USD 1 per order, USD 0.005 per share and 4 bp of traded value. These are alternatives, not additive charges.

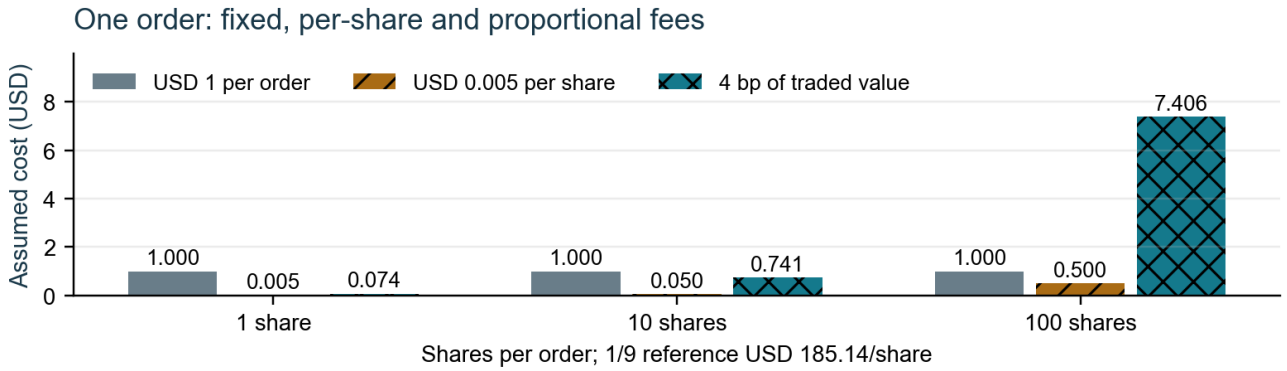
q is shares, C is price per share and F is cost in USD. Ten shares trade about USD 1,851.40; the three costs are USD 1.00, 0.05 and 0.74056. These are hypothetical fee schedules, not a borrowed ten-share order in the main account.

$$F_{\text{fixed}} = 1, \quad F_{\text{share}} = 0.005q, \quad F_{\text{value}} = 0.0004qC$$

Predict: which fee remains USD 1 when shares increase?

```
counts = np.array([1, 10, 100])
pd.DataFrame({"Order shares": counts, "Per-order fee (USD)": 1.0,
    "Per-share fee (USD)": counts*0.005,
    "Proportional fee (USD)": (counts*price*0.0004).round(5)})
```

Order shares	Per-order fee (USD)	Per-share fee (USD)	Proportional fee (USD)
1	1.0	0.005	0.07406
10	1.0	0.05	0.74056
100	1.0	0.5	7.4056



The fixed charge remains USD 1 per order. Per-share cost rises with shares; proportional cost rises with traded value. Do not add the three alternatives.

Discuss

If ten shares are split into ten one-share orders, how much does the USD 1 per-order schedule charge?

Sources: [2], [5].

ISSUE 04

At USD 185.14, how much extra would an immediate purchase cost?

An immediate purchase normally faces the ask; a sale faces the bid. Their difference is the bid-ask spread. Execution at the midpoint is not guaranteed.

Use the observed 1/9 Close, USD 185.14/share, as a reference. Assume a 4 bp full spread: bid 2 bp below and ask 2 bp above it. The reference is centered only for this calculation; the quotes are assumptions, not observed quotes.

$$185.14(2/10000) \approx 0.03703 \text{ USD/share}$$

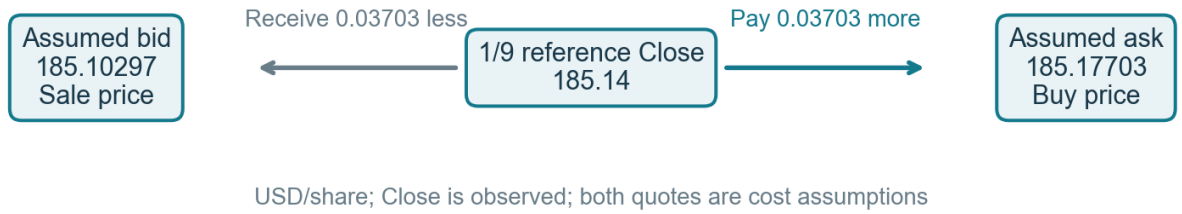
An assumed immediate purchase of one share costs about 185.14 + 0.03703 = USD 185.17703: about 3.70 cents above the reference, before commission or extra slippage.

Predict: how many dollars above the reference is the assumed purchase price?

```
reference_price = close["2024-01-09"]
half_spread = reference_price*2/10000
assumed_bid = reference_price-half_spread
assumed_ask = reference_price+half_spread
pd.DataFrame({"Price role": ["Assumed bid: sell", "Reference Close", "Assumed ask: buy"],
    "USD/share": np.round([assumed_bid, reference_price, assumed_ask], 5)})
```

Price role	USD/share
Assumed bid: sell	185.10297
Reference Close	185.14
Assumed ask: buy	185.17703

Same reference: buy above it and sell below it by half the assumed spread



USD/share; Close is observed; both quotes are cost assumptions

2 bp per direction is about 3.70 cents, not a full 4 bp spread for both buy and sell. Actual orders require contemporaneous quotes.

Discuss

At these unchanged assumed quotes, buy one share and immediately sell it. How many dollars are lost to the spread alone?

Sources: [3], [4].

ISSUE 05

When is the spread counted twice?

If the full extra amount from reference to fill is already included, do not subtract its spread component again. Follow the price of the same one-share purchase.

Reference: USD 185.14/share; assumed ask: 185.17703. Assume a further 1 bp beyond the ask, still based on 185.14: an extra 0.01851 gives a fill of about 185.19554. All quotes, fills and rates here are assumptions, not observations.

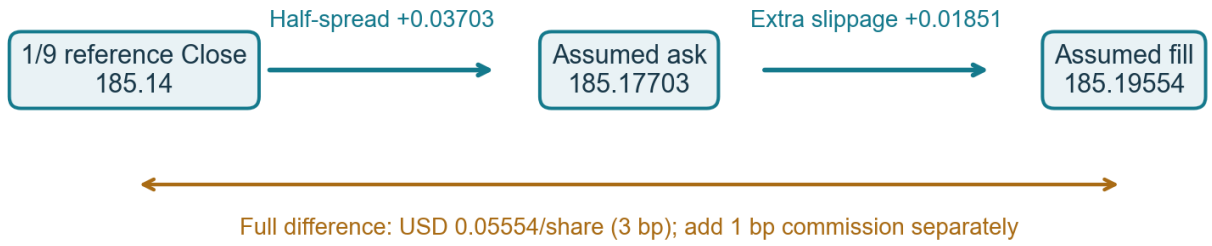
The USD 0.05554 reference-to-fill difference includes 0.03703 half-spread and 0.01851 extra slippage: 2 + 1 = 3 bp. Add 1 bp commission for 4 bp total. Adding half-spread again overcharges by 2 bp.

Predict: which row charges the same spread again?

```
extra_slippage = reference_price*1/10000
assumed_fill = assumed_ask+extra_slippage
rates = np.array([4, 4, 6])
pd.DataFrame({"Calculation": ["1+2+1", "1+3", "1+3+2: double count"],
    "Rate (bp)": rates,
    "One-share cost (USD)": (price*rates/10000).round(5)})
```

Calculation	Rate (bp)	One-share cost (USD)
1+2+1	4	0.07406
1+3	4	0.07406
1+3+2: double count	6	0.11108

One share: the reference-to-fill difference already includes half-spread



When using fill prices, check which price differences are already included; do not add them again. Actual slippage can also be favorable.

Discuss

If the fill equals this assumed ask with no extra slippage, how many dollars above the reference does one share cost, including 1 bp commission?

Sources: [3], [4], [5].

ISSUE 06

Record the purchase cost on the 9th or the 10th?

Buying at the 1/9 Close incurs cost that day. The first holding return ends on the 10th. Do not date both events identically.

Decide to hold stock on the 8th and buy at the 9th Close, 185.14. Hold from the 9th to the 10th, then sell at the 10th Close. held is the stock weight over the ending interval; after is the post-trade weight. Here 1 means all stock, 0 cash, not shares.

Predict: which date incurs purchase cost before earning a holding return?

```
held = signal.shift(2).fillna(0)
after = signal.shift(1).fillna(0)
after.iloc[-1] = held.iloc[-1]
change = after-held
turnover = change.abs()
actions = {1: "Buy", 0: "No trade", -1: "Sell"}
dates = ["2024-01-08", "2024-01-09", "2024-01-10"]
pd.DataFrame({"Observation date": dates,
    "Held weight": held.loc[dates].astype(int).values,
    "Post-trade weight": after.loc[dates].astype(int).values,
    "Action": [actions[int(change[d])] for d in dates]})
```

Observation date	Held weight	Post-trade weight	Action
2024-01-08	0	0	No trade
2024-01-09	0	1	Buy
2024-01-10	1	0	Sell

On the 9th, 0 to 1 means buy; on the 10th, 1 to 0 means sell. after uses the previous signal: decide at the previous Close, execute at this Close. held is one row earlier. Set final after equal to held so 6/28 only values the account. The execution assumption determines these shifts.

One-way turnover here is the absolute change in stock weight across execution: 0→1 or 1→0 gives 1; no trade gives 0. It determines whether to charge 4 bp that day, not a number of shares.

Pay for buying on 1/9; the first holding return ends on 1/10



Record the purchase cost on 1/9 even before earning a stock return. On 1/10, value the holding return and record the separate sale cost.

Discuss

Why must the 1/9 purchase cost not be postponed to the 10th?

Sources: [5], [6].

ISSUE 07

Does a purchase and a sale each incur a cost?

Continuing to hold is not buying again every day. A completed round trip does include two charged directions: purchase and sale.

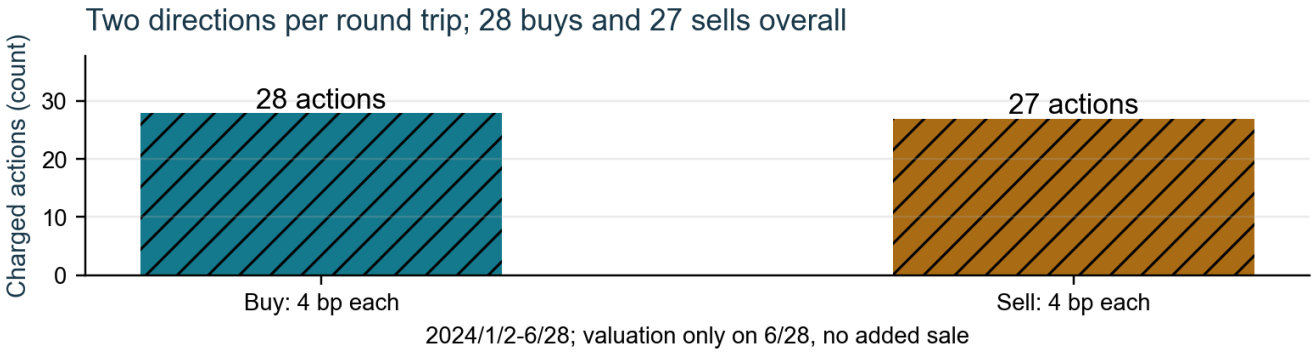
Predict: for buying on 1/9 and selling on 1/10, is 4 bp charged once or twice?

```
trades = pd.DataFrame({"Action": change, "Turnover": turnover})
trades = trades.loc[turnover.gt(0)].copy()
trades["Direction"] = np.where(trades["Action"].gt(0), "Buy", "Sell")
shown = trades.iloc[:2]
pd.DataFrame({"Trade date": shown.index, "Direction": shown["Direction"].values,
    "Cost (bp)": 4*shown["Turnover"].values})
```

Trade date	Direction	Cost (bp)
2024-01-09	Buy	4.0
2024-01-10	Sell	4.0

Charge 4 bp at each trade, but the bases differ. Calculate dollar costs separately (Issue 8), not 8 bp times one common amount.

There are 28 buys, 27 sells and 55 charged actions. On 6/28 stock is valued but not sold, so the final position is not a completed round trip. Do not add a sale merely to equalize buy and sell counts.



The first 1/9 buy and 1/10 sale each cost 4 bp. The period has 55 charged actions; do not add a final sale to complete the last round trip.

Discuss

If all stock is instead sold for cash on 6/28, which additional cost is incurred?

Sources: [2], [5].

ISSUE 08

First round trip: how much of USD 1,000 remains?

Follow capital, holding P/L and both costs from the 1/9 buy to the 1/10 sale.

Before buying on 1/9, there is USD 1,000. Approximate cost is 0.40, leaving USD 999.60. The price rises from 185.14 to 186.19 at the 10th Close: about +0.567140% per share.

$$999.60\left(\frac{186.19}{185.14}-1\right)\approx 5.669 \text{ USD}$$

To subtract costs directly from returns, the main model charges on opening account value. Approximate the sale cost as 999.60 × 0.0004 = USD 0.39984; then compare a charge based on sale value below.

Predict: with no holding P/L on the purchase day, can ending wealth still be below 1,000? All amounts are USD.

```
first_buy_fee = 1000 * 0.0004
first_open = 1000 - first_buy_fee
first_gain = first_open * (close["2024-01-10"] / close["2024-01-09"] - 1)
first_exit_fee = first_open * 0.0004
first_end = first_open + first_gain - first_exit_fee
pd.DataFrame({"Execution date": ["2024-01-09", "2024-01-10"],
    "Opening value": [f"{1000:.5f}", f"{first_open:.5f}"],
    "Holding P/L": ["0.00000", f"{first_gain:.5f}"],
    "Approximate cost": [f"{first_buy_fee:.5f}", f"{first_exit_fee:.5f}"],
    "Ending value": [f"{first_open:.5f}", f"{first_end:.5f}"]})
```

Execution date	Opening value	Holding P/L	Approximate cost	Ending value
2024-01-09	1000.00000	0.00000	0.40000	999.60000
2024-01-10	999.60000	5.66913	0.39984	1004.86929

Value before sale is USD 1005.26913. Charging 4 bp on that amount costs about 0.40211, or USD 0.00227 more than the main approximation. The price rise made sale value exceed opening wealth of 999.60.

Buy 1/9, sell 1/10: linear cost approximation using opening wealth



Sale-value charge: USD 0.40211; this diagram keeps the opening-value approximation

Same rate, different base: dollar costs differ. Table profit retains source-price precision.

Later issues keep the main approximation. For an actual bill, identify whether fees use traded value, shares or a fixed per-order amount; 4 bp alone is not enough.

Discuss

If only this sale cost changes to 4 bp of USD 1005.26913 traded value, how much remains after selling?

Source: [5].

ISSUE 09

Why subtract 4 basis points to obtain the net return?

Divide dollar cost by the same opening wealth before subtracting it from return. Dollars cannot be subtracted directly from a percentage.

w is opening stock weight; r is decimal stock return; c=0.0004. τ is one-way turnover, F cost (USD), and W opening wealth (USD). w and τ are 0 or 1. gross/net mean account returns before/after cost, in decimals.

$$r_{\text{net}} = wr - c\tau, \qquad F/W = c\tau$$

On 1/10, cost 0.39984 / opening wealth 999.60 = 0.04%. Holding return 0.567140% minus 0.04% gives net return 0.527140%. Directly subtracting 4 bp works because the cost uses that same opening value.

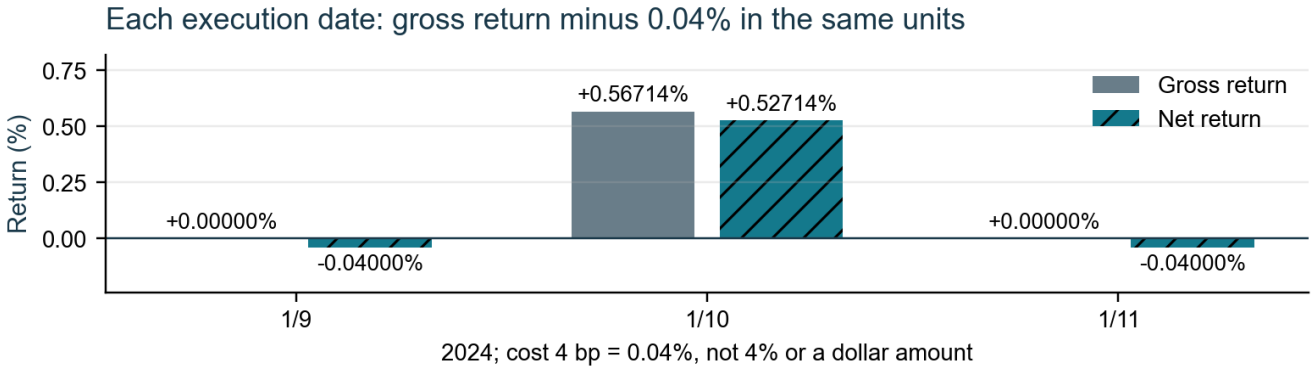
$$\frac{5.66913 - 0.39984}{999.60} \approx 0.00527140 = 0.527140\%$$

Predict: why is the 9th net return negative before earning a stock return?

```
asset_return = close.pct_change(fill_method=None)
gross = asset_return.where(held.eq(1), 0.0)
cost_rate = 0.0004
net = gross-cost_rate*turnover
dates = ["2024-01-09", "2024-01-10", "2024-01-11"]
pd.DataFrame({"Observation date": dates,
    "Holding return (%)": (100*gross.loc[dates]).round(5).values,
    "Cost (bp)": (4*turnover.loc[dates]).values,
    "Net return (%)": (100*net.loc[dates]).round(5).values})
```

Observation date	Holding return (%)	Cost (bp)	Net return (%)
2024-01-09	0.0	4.0	-0.04
2024-01-10	0.56714	4.0	0.52714
2024-01-11	0.0	4.0	-0.04

If charged on 1/10 sale value, cost is about USD 0.40211. Still divide by opening wealth 999.60 before subtracting from the interval return. 4 bp of sale value is not exactly 4 bp of opening wealth.



Discuss

If cost becomes USD 0.40211, divide it by opening wealth 999.60. Is net return still 0.527140%?

Sources: [5], [6].

ISSUE 10

Why does the terminal wealth gap differ from total fees?

Start with the first round trip. Paying the buy cost leaves less money invested, changing subsequent stock profit or loss. Summing fees and deducting them only at the end misses this effect.

After the 1/9 purchase, the gross account is USD 1,000 and the net account 999.60: a 0.40 gap. Both hold stock the next day. As price rises from 185.14 to 186.19, that gap grows to about 0.40227. Add the 0.39984 sale cost for a gap of about 0.80211.

$$0.40\left(\frac{186.19}{185.14}\right) + 0.39984 \approx 0.80211 \text{ USD}$$

The two fees total only 0.40 + 0.39984 = USD 0.79984. The extra 0.00227 comes from the difference in stock investment created by the first fee, not a third charge.

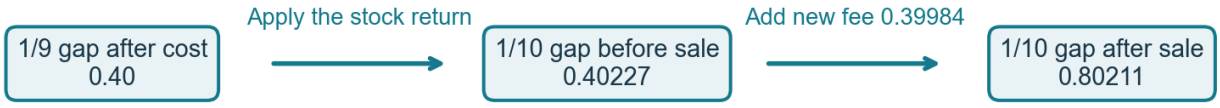
Predict: will the 1/10 account gap equal the two fees totaling USD 0.79984?

```
gross_wealth = 1000*(1+gross).cumprod(skipna=False)
net_wealth = 1000*(1+net).cumprod(skipna=False)
opening = net_wealth.shift(1).fillna(1000)
fees = opening*cost_rate*turnover
cumulative_fees = fees.cumsum()
wealth_gap = gross_wealth-net_wealth
dates = ["2024-01-09", "2024-01-10", "2024-06-28"]
pd.DataFrame({"Observation date": dates,
    "Fees to date (USD)": cumulative_fees.loc[dates].round(5).values,
    "Account gap (USD)": wealth_gap.loc[dates].round(5).values})
```

Observation date	Fees to date (USD)	Account gap (USD)
2024-01-09	0.4	0.4
2024-01-10	0.79984	0.80211
2024-06-28	22.04634	24.62569

On 6/28, gross wealth is 1134.06 and net wealth 1109.43: a USD 24.63 gap, versus 22.05 in nominal fees. Subsequent returns accumulate these differences; stock declines can also make the gap smaller than total fees.

2024/1/9 to 1/10: the existing gap earns the stock return, then add the new fee



Amounts in USD; the two nominal fees total 0.79984

Money no longer invested also misses stock gains or losses. Charging each trade and deducting all fees at the end therefore give different amounts.

Discuss

If USD 22.05 in fees is deducted only from gross terminal wealth 1134.06 on 6/28, how much does this overstate the actual net wealth 1109.43?

Sources: [2], [5].

ISSUE 11

How can USD 1,000 cover both stock and purchase cost?

USD 1,000 must pay for both stock and the purchase cost. Spending all 1,000 on stock leaves no cash for fees.

The 1/9 reference Close C is about USD 185.14/share; q is shares. Assume c = 0.0004 charged on stock traded value. First solve stock value + fee = available cash; do not divide all cash by price before reserving the fee.

$$qC + cqC = 1000 \quad \Longrightarrow \quad q = \frac{1000}{C(1+c)}$$

$$q \approx \frac{1000}{185.14(1.0004)} \approx 5.399158 \text{ shares}$$

Predict: which row needs more than USD 1,000?

```
q_without = 1000/price
q_reserved = 1000/(price*(1+cost_rate))
counts = np.array([q_without, q_reserved])
pd.DataFrame({"Purchase budget": ["No cost allowance", "Cost reserved"],
    "Shares": counts.round(6),
    "Cash required (USD)": (counts*price*(1+cost_rate)).round(4)})
```

Purchase budget	Shares	Cash required (USD)
No cost allowance	5.401318	1000.4
Cost reserved	5.399158	1000.0

Without reserving fees, USD 1000.40 is needed: 0.40 too much. With the reserve, stock 999.60016 plus fee 0.39984 totals 1000, buying about 5.399158 shares. This maximum assumes unrestricted fractional shares.

The main approximation first deducts 1000 × 4 bp = 0.40; this budget charges the actual stock amount. The difference is under one cent, but the bases differ. Real orders must reserve fees under traded-value and share constraints. Later issues keep the main approximation.

Exact purchase budget: stock value plus cost equals required cash



All amounts are USD. Exact stock value is 1,000÷1.0004; stock plus cost equals 1,000.

Discuss

If only whole shares are allowed, which is affordable with USD 1,000 at 185.14/share and a 4 bp fee: 5 shares or 6?

Source: [5].

ISSUE 12

How does buying one close later change the first holding return?

Keep the 1/8 signal and change only purchase delay. First compare the actual holding intervals' price changes.

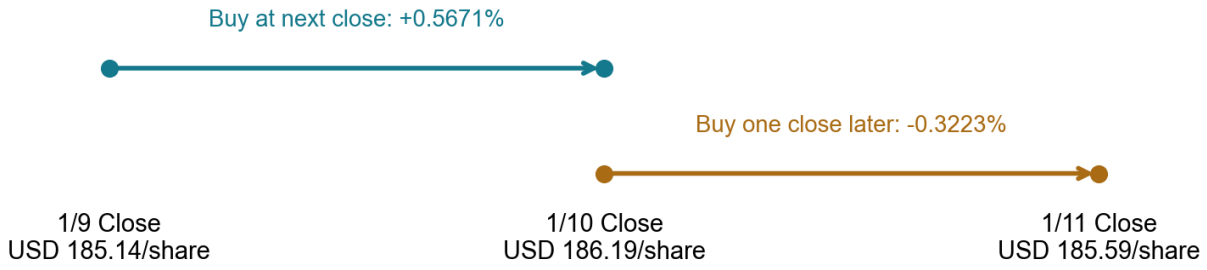
Execution plan	Buy date / price	First end / price
Next close	1/9：185.14	1/10：186.19
One close later	1/10：186.19	1/11：185.59

Prices are USD/share. Fix capital and signal; change only execution delay. The original first return is about +0.5671%; the delayed first return is about -0.3223%. This compares stock prices only, without costs or later trades.

$$\frac{186.19}{185.14} - 1 \approx 0.5671\%, \qquad \frac{185.59}{186.19} - 1 \approx -0.3223\%$$

Delaying execution changes both holding intervals and cost dates. It is not merely moving the result column, or measuring actual slippage from daily data.

Same 1/8 decision: delay changes the first holding dates



The original 1/9 to 1/10 return is positive; the delayed 1/10 to 1/11 return is negative. Later trades and costs are not yet included.

Discuss

If buying only at the 1/10 Close, can the account earn the 1/9 to 1/10 return of +0.5671%?

Sources: [4], [6].

ISSUE 13

Can the first worse return determine the entire delayed account?

Shift every holding interval and cost date together to compare full accounts over the same 1/2 to 6/28 period.

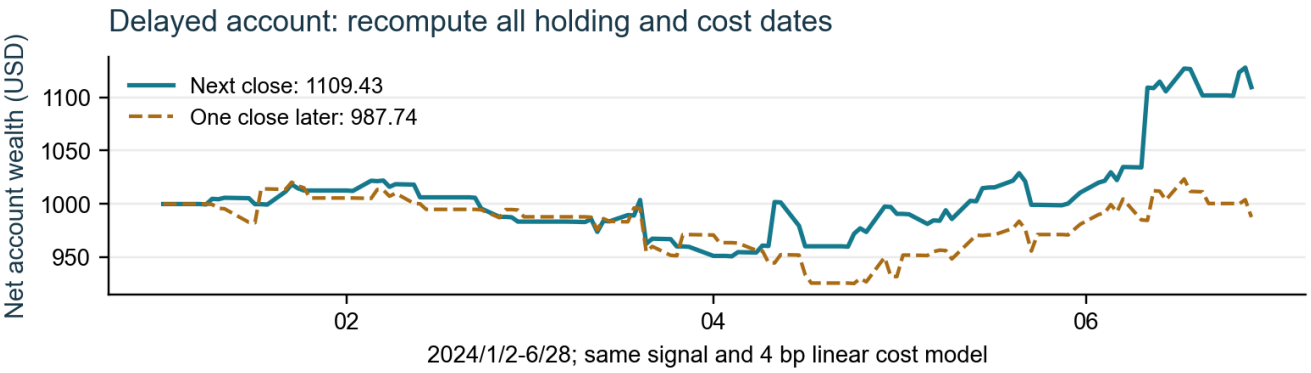
The original account follows the 1/8 signal to buy at the 1/9 Close. The delayed account buys on 1/10; its first return ends on 1/11. Its post-trade state therefore looks back two rows and its holding state three. Both charge 4 bp per direction; the final date is valuation only.

Predict: is a worse first interval enough to calculate the complete terminal gap?

```
delayed_held = signal.shift(3).fillna(0)
delayed_after = signal.shift(2).fillna(0)
delayed_after.iloc[-1] = delayed_held.iloc[-1]
delayed_turnover = (delayed_after-delayed_held).abs()
delayed_gross = asset_return.where(delayed_held.eq(1), 0.0)
delayed_net = delayed_gross-cost_rate*delayed_turnover
delayed_wealth = 1000*(1+delayed_net).cumprod(skipna=False)
pd.DataFrame({"Execution plan": ["Next close", "One close later"],
    "First purchase": ["2024-01-09", "2024-01-10"],
    "Ending net wealth (USD)": [round(net_wealth.iloc[-1], 2),
                        round(delayed_wealth.iloc[-1], 2)]})
```

Execution plan	First purchase	Ending net wealth (USD)
Next close	2024-01-09	1109.43
One close later	2024-01-10	987.74

Terminal wealth is USD 1109.43 originally and 987.74 with delay. Delay also changes every later entry and exit interval. The first interval cannot determine all subsequent results, and this example does not show that every delay reduces return.



Delay changes every later entry and exit, not just the first. These historical paths do not show that all delays produce worse results.

Discuss

Why must cost dates change along with holding dates in the delayed account?

Sources: [4], [6].

ISSUE 14

Do fewer trades necessarily leave more money?

Adopting a new decision every five observations changes both trading frequency and holding intervals. Separate these effects.

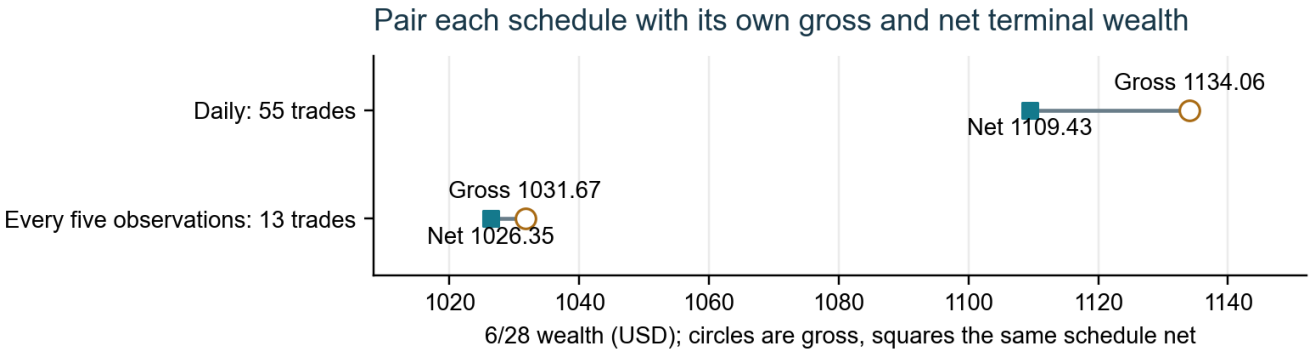
Still calculate the one-day signal daily. Adopt a new decision every five observations from 1/2: 1/2, 1/9, 1/17, and so on; retain it between dates. This is not a five-day return signal or a calendar-week trading schedule.

Predict: with fewer trades, must gross terminal wealth stay the same?

```
scheduled_signal = signal.iloc[::5].reindex(signal.index).ffill()
scheduled_held = scheduled_signal.shift(2).fillna(0)
scheduled_after = scheduled_signal.shift(1).fillna(0)
scheduled_after.iloc[-1] = scheduled_held.iloc[-1]
scheduled_turnover = (scheduled_after-scheduled_held).abs()
scheduled_gross = asset_return.where(scheduled_held.eq(1), 0.0)
scheduled_net = scheduled_gross-cost_rate*scheduled_turnover
pd.DataFrame({"Decision schedule": ["Daily", "Every five observations"],
    "Trade count": [int(turnover.sum()), int(scheduled_turnover.sum())],
    "Gross terminal value (USD)": [round(gross_wealth.iloc[-1], 2),
                      round(1000*(1+scheduled_gross).prod(), 2)],
    "Net terminal value (USD)": [round(net_wealth.iloc[-1], 2),
                      round(1000*(1+scheduled_net).prod(), 2)]})
```

Decision schedule	Trade count	Gross terminal value (USD)	Net terminal value (USD)
Daily	55	1134.06	1109.43
Every five observations	13	1031.67	1026.35

Daily decisions produce 55 trades; every five observations produces 13. The latter ends at gross 1031.67 and net 1026.35, both below daily results. Changed gross wealth shows changed holding periods; the entire difference cannot be attributed to costs.



Fewer trades can still leave less net wealth. Compare gross versus net within a row; a difference across schedules is not solely a cost effect.

Discuss

Which two table values isolate costs without changing holding periods?

Sources: [5], [6].

ISSUE 15

What happens to terminal wealth when only the cost rate rises?

Fix prices, signal, execution, holdings and endpoint before attributing terminal changes to the cost rate.

With positive growth factors and nonnegative costs, increasing the rate cannot increase an interval's factor or their compounded terminal value.

W is wealth in USD; T is 6/28. b is the rate in bp, r_{gross} is gross account return, τ is one-way turnover, and t is each interval's ending date. At the first 1/9 purchase with 10 bp, 1000×(1-0.001)=USD 999.

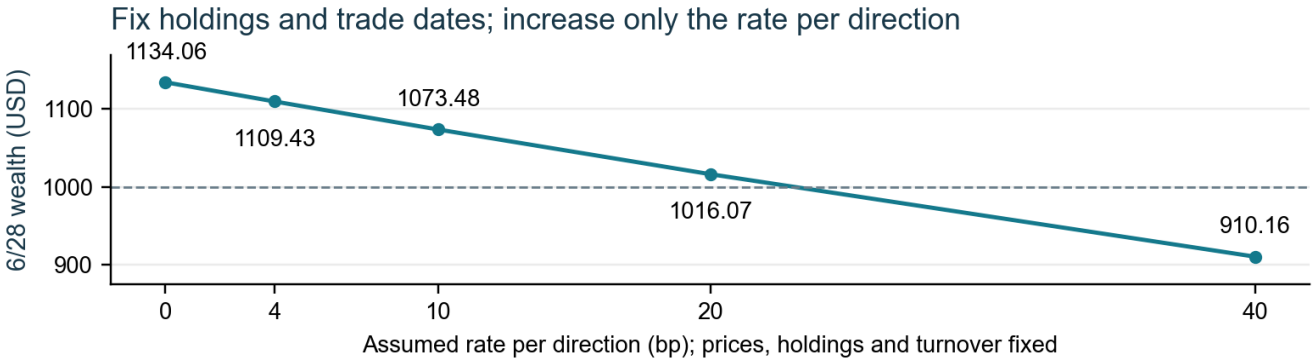
$$W_T(b) = 1000\prod_t (1 + r_{\text{gross},\,t} - \tau_t b/10000)$$

Predict: with other conditions fixed, can terminal wealth at 40 bp exceed that at 4 bp?

```
sensitivity = []
for bp in [0, 4, 10, 20, 40]:
    factors = 1+gross-turnover*bp/10000
    assert factors.gt(0).all()
    sensitivity.append([bp, 1000*factors.prod()])
pd.DataFrame(sensitivity,
    columns=["Rate per direction (bp)", "Terminal wealth (USD)"]).round(2)
```

Rate per direction (bp)	Terminal wealth (USD)
0	1134.06
4	1109.43
10	1073.48
20	1016.07
40	910.16

At 0 bp wealth is 1134.06; at 4 bp, 1109.43; at 40 bp, 910.16. This is sensitivity to specified rates, not actual broker fees identified from daily prices.



A higher rate lowers terminal wealth on this fixed path. The curve does not estimate actual rates from market data.

Discuss

If terminal wealth at 40 bp is higher instead, which holding condition should be checked first?

Sources: [2], [5].

ISSUE 16

Which cost rate eliminates this historical profit?

Find the assumed rate that brings terminal wealth back to USD 1,000, not gross profit divided by trade count.

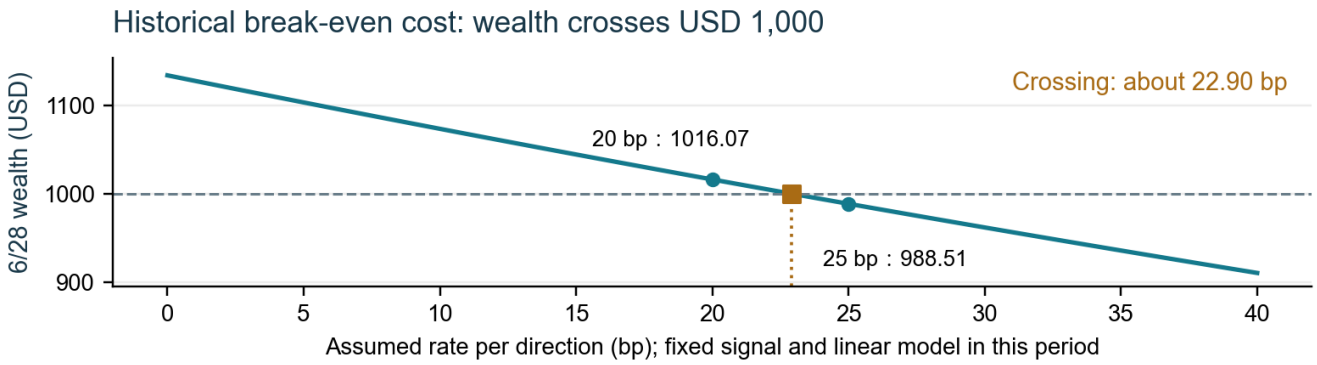
At 0 bp, wealth 1134.06 exceeds 1000; at 40 bp, 910.16 is below 1000. Along this fixed path, wealth decreases continuously with cost, giving one crossing. Try 20 bp: 1016.07. Keep 20 to 40, then try 30 bp; repeatedly narrow the interval.

Predict: 25 bp exceeds 20 bp. Which rate still leaves this historical account above USD 1,000?

```
lo, hi = 0.0, 40.0
assert 1000*(1+gross-turnover*lo/10000).prod() > 1000
assert 1000*(1+gross-turnover*hi/10000).prod() < 1000
for _ in range(50):
    mid = (lo+hi)/2
    value = 1000*(1+gross-turnover*mid/10000).prod()
    if value > 1000:
        lo = mid
    else:
        hi = mid
break_even = (lo+hi)/2
comparison_bp = [20.0, break_even, 25.0]
comparison_end = [1000*(1+gross-turnover*b/10000).prod()
                  for b in comparison_bp]
pd.DataFrame({"Rate per direction (bp)": np.round(comparison_bp, 2),
    "Terminal wealth (USD)": np.round(comparison_end, 2)})
```

Rate per direction (bp)	Terminal wealth (USD)
20.0	1016.07
22.9	1000.0
25.0	988.51

The crossing is about 22.90 bp. At 20 bp, USD 1016.07 remains; at 25 bp, only 988.51. This tests whether historical profit depends on an unrealistically low assumed cost. Actual fees need separate evidence, and future returns need not repeat.



At 25 bp this historical account ends below its initial capital; 22.90 bp does not guarantee future profit.

Discuss

If the appropriate cost is 25 bp per direction, does the USD 109.43 profit calculated at 4 bp remain? Answer using the table.

Sources: [2], [5].

ISSUE 17

How do cash, buy-and-hold and the trading rule compare?

Fix initial USD 1,000 and the 1/2 to 6/28 period. Each holding approach pays its own required trading costs.

Cash holds no stock, trades none and earns no interest. Buy-and-hold buys at the 1/2 Close of USD 185.64/share and pays 4 bp once; the one-day rule pays 55 times. Both value stock at the 6/28 Close of USD 210.62/share without a final sale.

BH denotes buy-and-hold in the formula; W is dollar wealth and T is 6/28. Initial stock value 999.60 still uses the same linear approximation.

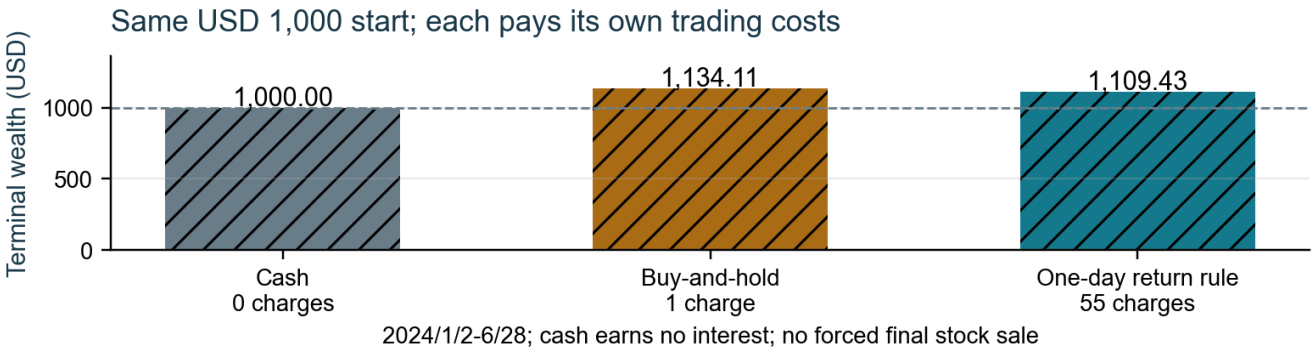
$$W_T^{\mathrm{BH}} = 1000(1-0.0004)\frac{210.62}{185.64} \approx 1134.11 \text{ USD}$$

Predict: is buy-and-hold also charged 4 bp every day?

```
buy_hold_wealth = 1000*(1-cost_rate)*close/close.iloc[0]
pd.DataFrame({"Method": ["Cash", "Buy-and-hold", "One-day return rule"],
    "Charge count": [0, 1, int(turnover.sum())],
    "Terminal wealth (USD)": [1000.0,
        round(buy_hold_wealth.iloc[-1], 2), round(net_wealth.iloc[-1], 2)]})
```

Method	Charge count	Terminal wealth (USD)
Cash	0	1000.0
Buy-and-hold	1	1134.11
One-day return rule	55	1109.43

Buy-and-hold ends at 1134.11 and the main rule at 1109.43. More trades do not imply more money in this period. The 1/2 buy-and-hold cost belongs in wealth, but is an initial-boundary cost, not one of the following 123 daily market returns. Week 7 separates them.



Daily valuation is not daily trading. Buy-and-hold pays only the initial purchase cost; the one-day rule pays for 55 actions.

Discuss

With no final buy-and-hold sale, how many times is this account charged?

Sources: [2], [5].

ISSUE 18

With a USD 1 minimum per order, does a small order still cost only 4bp?

A USD 1 minimum per order can make cost exceed 4bp of traded value.

Separately assume commission of 4bp of traded value with a USD 1 minimum per order. Exclude spread, slippage and tax. This is neither a broker quote nor the main account's composite 4bp cost. Use the 1/9 AAPL Close, USD 185.14/share, without changing the main account.

q is order shares, C is USD/share, and F is commission in USD. max selects the larger amount. Divide F by stock traded value qC, then multiply by 10,000 to express the rate in bp.

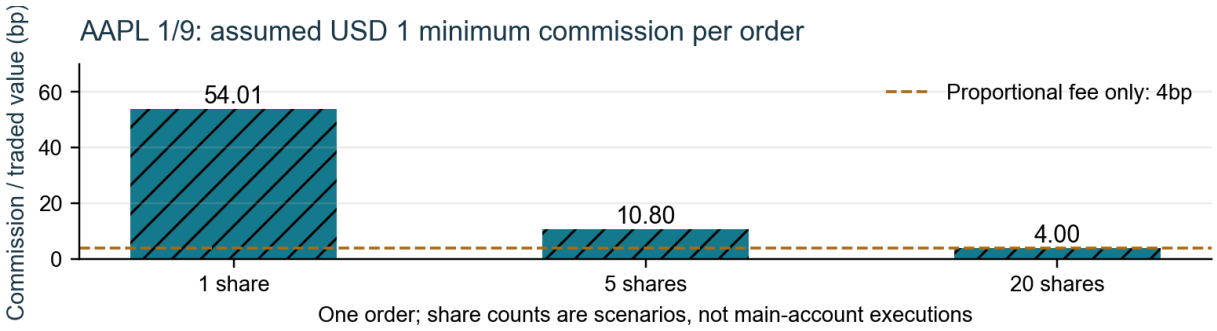
$$F = \max(1, 0.0004qC), \qquad \text{fee in bp} = \frac{F}{qC}10000$$

For one share, 0.0004×185.14≈USD 0.07406, replaced by the minimum 1. Then 1÷185.14×10,000≈54.01bp. Share counts only test the fee schedule; 20 shares are not a main-account order.

Predict: under this same fee schedule, do one and five shares have the same cost rate in bp?

```
order_units = np.array([1, 5, 20])
order_amount = order_units*price
order_fee = np.maximum(1.0, .0004*order_amount)
order_bp = order_fee/order_amount*10000
pd.DataFrame({"Shares": order_units, "Traded value (USD)": order_amount.round(2),
    "Commission (USD)": order_fee.round(5), "Cost rate (bp)": order_bp.round(2)})
```

Shares	Traded value (USD)	Commission (USD)	Cost rate (bp)
1	185.14	1.0	54.01
5	925.7	1.0	10.8
20	3702.8	1.48112	4.0



Five shares still cost USD 1, or 10.80bp; 20 shares pay 4bp. The stock-value threshold is 1÷0.0004=USD 2,500, requiring USD 2,501 cash including cost. Splitting small orders repeats the minimum charge.

Discuss

With a separate USD 1,000 cash budget, what is the maximum ideal fractional purchase in one order under this fee schedule? Why is Issue 11's proportional-only budget inapplicable?

Sources: [2], [5].

References

Price sources and reproduction conditions

[1] yfinance: historical price fields

[2] Investor.gov: understanding fees

[3] Investor.gov: ask price and bid-ask spread

[4] Investor.gov: executing an order

[5] Lobo, Fazel and Boyd: proportional and fixed transaction costs

[6] pandas: shifting observations

AAPL daily data: 2024/1/2 to 6/28, 124 observations from the Yahoo/yfinance snapshot retained on 2026/9/14. Trading rules, execution and costs are teaching assumptions, not actual execution records.

Prices display 2 decimal places; calculations retain source precision. Reproduce with the recorded data, without replacing results by a new download.

Python 3.12.14；numpy 2.2.6；pandas 2.2.3；matplotlib 3.10.8；yfinance 1.2.0。

Week 7 continues the original composite 4bp linear-approximation account, USD 1109.43, not Issue 18's minimum commission schedule.